



Rational Decision Making

Examples and Notes



Example

Utility Theory: MEU calculations with Bayesian Updating



Example: Panda's Lunch Choice

Partially Observable Environments

- Setup

- Panda Lulu chooses between Bamboo Grove and Berry Bush, but can't see the grove from her location.
- Actions: Bamboo Grove (risky) vs Berry Bush (sure 4).
- Prior over grove state:
 $P(B) = 0.5$ (bamboo present), $P(\neg B) = 0.5$ (empty).
- Utilities if Lulu goes to the grove: $U = 10$ if B ; $U = 0$ if $\neg B$.

- Observation:

- Lulu hears rustling near the grove. Observation $O \in \{\text{rustling}, \text{no rustling}\}$ before choosing. This could mean:
- Wind in the leaves — bamboo still there, happens 20% of the time if bamboo is present
- Monkeys eating bamboo — bamboo already gone, happens 60% of the time if bamboo is absent:
- Likelihoods (rustling more likely when grove is empty—monkeys):
 $P(O = \text{rustling} \mid B) = 0.2$, $P(O = \text{rustling} \mid \neg B) = 0.6$.

- Takeaways:

- Belief updating (Bayes) $\rightarrow$ new posteriors $\rightarrow$ new EUs $\rightarrow$ new choice.
- Same prior and utilities can yield different optimal actions depending on the observation.

1) Posterior after hearing rustling

- Calculations:

- $$P(B | O) = \frac{P(O|B)P(B)}{P(O|B)P(B)+P(O|\neg B)P(\neg B)} = \frac{0.2 \cdot 0.5}{0.2 \cdot 0.5 + 0.6 \cdot 0.5} = \frac{0.2}{0.2 + 0.6} = 0.25$$

- Expected utilities:

- $EU(\text{Grove} | O) = 0.25 \cdot 10 + 0.75 \cdot 0 = 2.5$

- $EU(\text{Berry}) = 4$

- Decision: Choose Berry Bush ($4 > 2.5$).

2) Posterior after hearing no rustling

- Calculations:

- Use complements: $P(\text{no} \mid B) = 0.8$, $P(\text{no} \mid \neg B) = 0.4$.

- $$P(B \mid \text{no}) = \frac{0.8 \cdot 0.5}{0.8 \cdot 0.5 + 0.4 \cdot 0.5} = \frac{0.8}{0.8 + 0.4} = \frac{2}{3} \approx 0.667$$

- Expected utilities:

- $$EU(\text{Grove} \mid \text{no}) = \frac{2}{3} \cdot 10 \approx 6.67$$

- $$EU(\text{Berry}) = 4$$

- Decision: Choose Bamboo Grove ($6.67 > 4$).



Constructing Utility Functions

Probability equivalence

Certainty equivalence



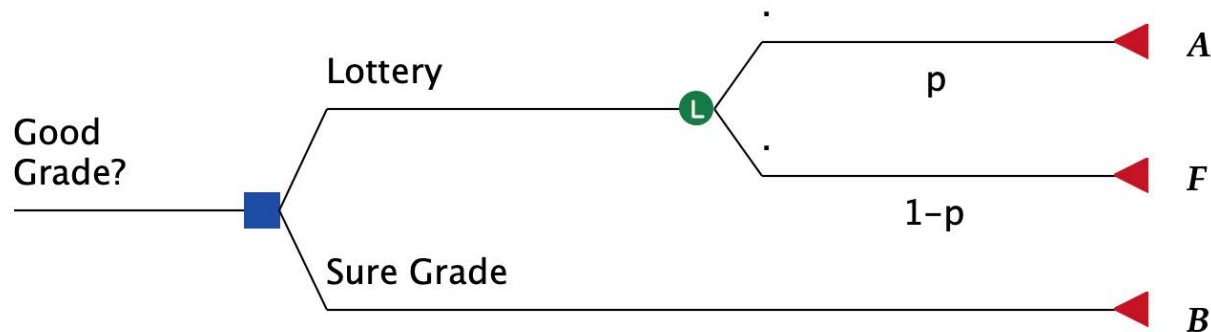
Preference Elicitation: Method 1

- Using probability equivalents
 - Present choices to agents; define utility function from observed responses
- Method:
 - Fix a scale; fix utilities of 2 outcomes to establish the scale
 - Worst: $u_{\perp}$; Best: $u_{\top}$
 - Normalized utilities: $u_{\perp} = 0; u_{\top} = 1$
 - Assess utility for prize S by asking the agent to choose between $U(S)$ and the lottery $[p, u_{\top}; (1 - p), u_{\perp}]$
 - Adjust p until the agent is indifferent between S and lottery

Example: Getting Good Grade

- Demonstration:

- Grade in CS4246/CS5446: $U(F) = 0$; $U(A) = 1$; sure grade: B, C, D
- What value of p would you trade *Lottery* for B ?



- What about C, D ?



Example: Pizza Party

- Demonstration:

- Alice's preference: $Pasta \succ Pizza \succ Salad$
- Bob's preference: $Salad \succ Pasta \succ Pizza$
- Charlie's preference: $Pizza \succ Pasta \succ Salad$

- What is the group's preference?

UNSURE! Compromise and other considerations are needed

- So?

- Group preference may be non-transitive
- Studied in social choice theory

Certainty Equivalent

- Certainty Equivalent (CE)

- Fixed amount of money equivalent to a given situation involving uncertainty
- Value agent accepts in lieu of the lottery

- Example:

- You are faced with the following lottery:

- | | | | | |
|--------|--------|------------------|-----|--|
| • Win | \$2000 | with probability | 0.5 | Expected Monetary Value
EMV = \$990 |
| • Lose | \$20 | with probability | 0.5 | |

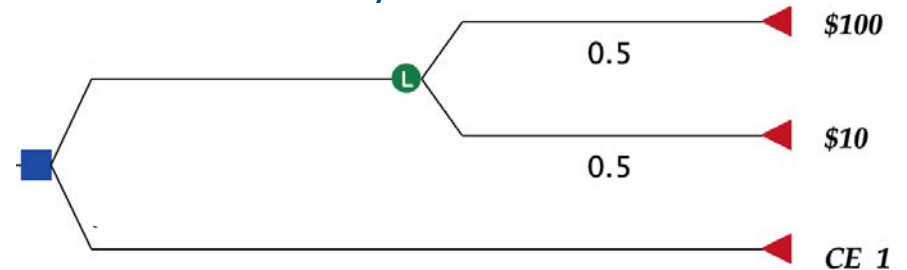
- How much are you willing to sell the lottery for?
- This amount, say \$ X , is the least that you would accept for the lottery, then the lottery must be **equivalent** in your mind to a sure \$ X

Preference Elicitation: Method 2

- Using certainty equivalents
 1. Determine two initial points on curve
 2. Arbitrarily assign utility values to the points
 3. Create reference lottery to determine CE
 4. Find third point on curve by formula:
 - $U(CE_1) = EU(Lottery)$
 5. Create additional reference lotteries
 6. Proceed as before until enough points are available to plot a curve

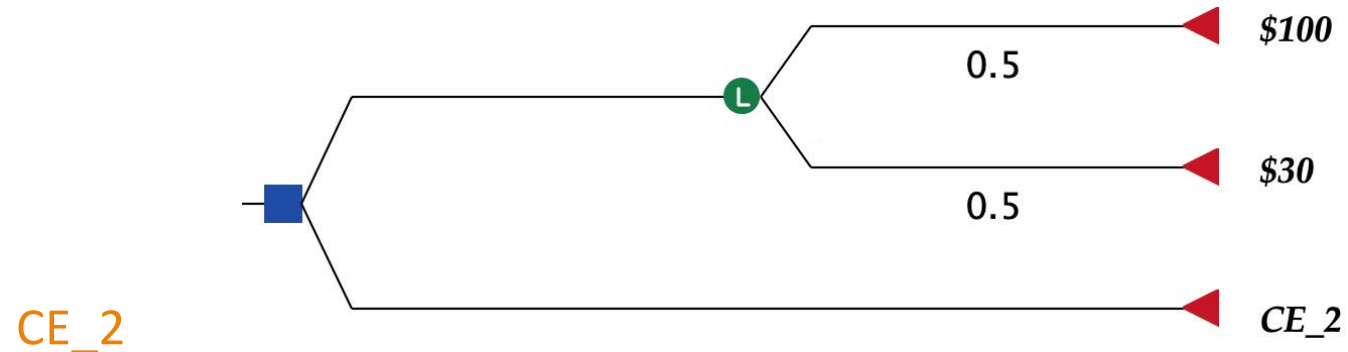
- Example:

1. Assume: worst: \$10, best: \$100
2. Set $U_{\perp}(10) = 0$; $U_{\top}(100) = 1$
3. Reference lottery

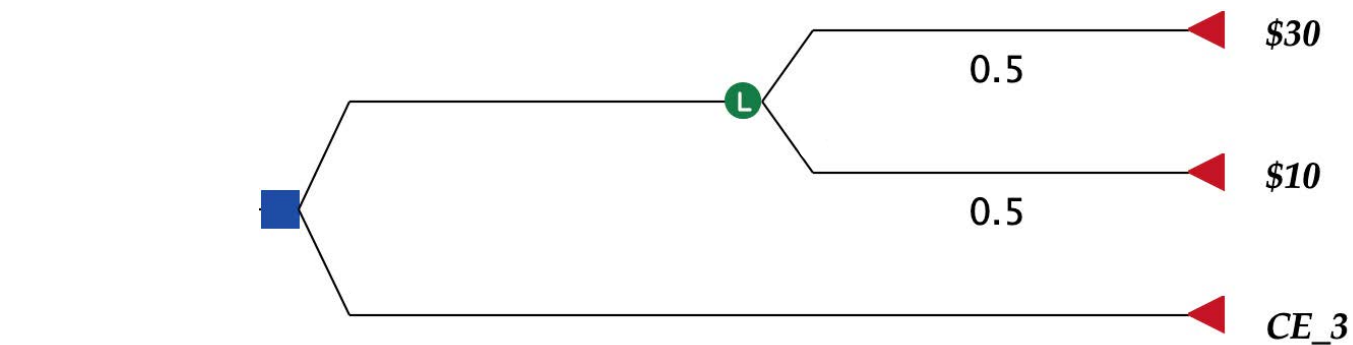


4. Assume: $CE_1 = 30$
5.
$$\begin{aligned} U(CE_1) &= U(30) \\ &= 0.5U(100) + 0.5U(10) \\ &= 0.5 \end{aligned}$$

Continue ...



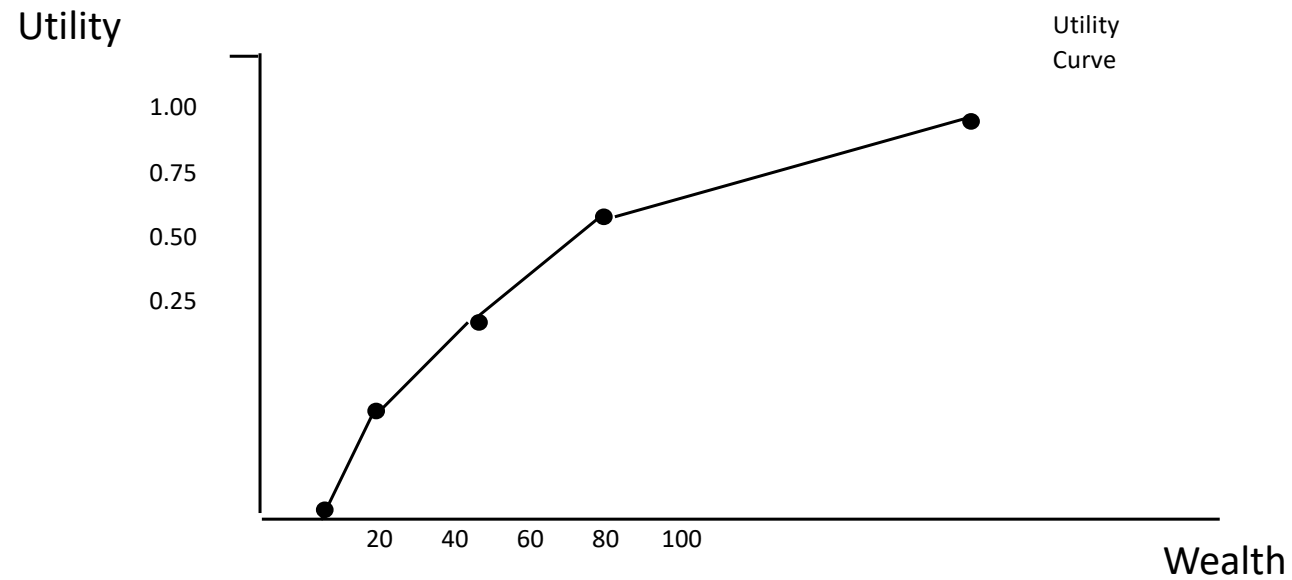
$$U(50) = 0.5U(100) + 0.5U(30) = 0.75$$



$$U(18) = 0.5U(30) + 0.5U(10) = 0.25$$



Plotting the Utility Function





Proof of Expected Utility Theorem



Expected Utility Theorem: Proof

- [Proof Sketch]:

- An orderable, transitive preference relation satisfies continuity and substitutability if and only if it admits an expected utility representation.

- Based on Levin, Jonathan. Choice Under Uncertainty. Lecture notes, 2006.

Access from:

<https://web.stanford.edu/~jdlevin/Econ%20202/Uncertainty.pdf>

Expected Utility Theorem: Proof

Proof:

We first argue that if U is an expected utility, it satisfies continuity and substitutability.

- Continuity: If $A \succ B \succ C$, then $U(A) > U(B) > U(C)$. Let $p = \frac{U(B) - U(C)}{U(A) - U(C)}$. Then $pU(A) + (1 - p)U(C) = U(B)$ showing that this value of p makes $[p, A; 1 - p, C] \sim B$.

Continuity:

$$A \succ B \succ C \Rightarrow \\ \exists p [p, A; 1 - p, C] \sim B.$$

Expected Utility Theorem: Proof

- Substitutability: If $A \sim B$, then $U(A) = U(B)$. Hence $pU(A) + (1 - p)U(C) = pU(B) + (1 - p)U(C)$.

Substitutability:

$$A \sim B \Rightarrow [p, A; 1 - p, C] \sim [p, B; 1 - p, C].$$

Now assume that the preference satisfies continuity and substitutability. We will construct an expected utility function U that represents the preference. Let $L_{\top}$ and $L_{\perp}$ represent the most and least preferred lotteries. Assume $L_{\top} \succ L_{\perp}$ (the result is trivial if $L_{\top} \sim L_{\perp}$).

Expected Utility Theorem: Proof

First we show **monotonicity**. If $1 > p > q > 0$, then

$$L_{\top} \succ [p, L_{\top}; 1 - p, L_{\perp}] \succ [q, L_{\top}; 1 - q, L_{\perp}] \succ L_{\perp}.$$

Aside: argument holds for any L_1, L_2 with $L_1 \succ L_2$.

- First inequality: write $L_{\top} = [p, L_{\top}; 1 - p, L_{\top}]$ and the inequality follows from substitutability as $L_{\top} \succ L_{\perp}$.
- Second inequality: write LHS as $[(p - q), L_{\top}; q, L_{\top}; 1 - p, L_{\perp}]$ and RHS as $[(p - q), L_{\perp}; q, L_{\top}; 1 - p, L_{\perp}]$, use substitutability.
- Third inequality: write RHS as $[q, L_{\perp}; 1 - q, L_{\perp}]$ and use substitutability.

Substitutability : $A \succ B \Rightarrow$

$$[p, A; 1 - p, C] \succ [p, B; 1 - p, C].$$

Monotonicity : $A \succ B \Rightarrow$

$$((p > q) \Leftrightarrow [p, A; 1 - p, B] \succ [q, A; 1 - q, B]).$$

Expected Utility Theorem: Proof

For any lottery L , there is a **equivalent lottery** $[p_L, L_\top; (1 - p_L), L_\perp]$ with a unique p_L such that $L \sim [p_L, L_\top; (1 - p_L), L_\perp]$. Existence follows from continuity while monotonicity implies uniqueness.

Continuity : $A \succ B \succ C \Rightarrow \exists p [p, A; 1 - p, C] \sim B$.

Monotonicity : $A \succ B \Rightarrow ((p > q) \Leftrightarrow [p, A; 1 - p, B] \succ [q, A; 1 - q, B])$.

We argue that $U(L) = p_L$ is an expected utility representation of the preference relation.

- First it gives the correct ordering:

$$\begin{aligned} L_1 &\succsim L_2 \\ \Leftrightarrow [p_{L_1}, L_\top; (1 - p_{L_1}), L_\perp] &\succsim [p_{L_2}, L_\top; (1 - p_{L_2}), L_\perp] \text{ (equiv lottery)} \\ \Leftrightarrow p_{L_1} &\geq p_{L_2} \text{ (monotonicity)} \\ \Leftrightarrow U(L_1) &\geq U(L_2) \text{ (defn of utility)} \end{aligned}$$

Expected Utility Theorem: Proof

- Also $U(L) = p_L$ is an expected utility, i.e. for any L_1, L_2 and α

$$U([\alpha, L_1; (1 - \alpha), L_2]) = \alpha U(L_1) + (1 - \alpha)U(L_2).$$

- We know that we can represent L_1 and L_2 by equivalent lotteries

$$L_1 \sim [U(L_1), L_{\top}; (1 - U(L_1)), L_{\perp}]$$

$$L_2 \sim [U(L_2), L_{\top}; (1 - U(L_2)), L_{\perp}]$$

- Using substitutability ($A \sim B \Rightarrow [p, A; 1 - p, C] \sim [p, B; 1 - p, C]$)

$$[\alpha, L_1; (1 - \alpha), L_2]$$

$$\sim [\alpha, [U(L_1), L_{\top}; (1 - U(L_1)), L_{\perp}]; (1 - \alpha), [U(L_2), L_{\top}; (1 - U(L_2)), L_{\perp}]]$$

$$\sim [\alpha U(L_1) + (1 - \alpha)U(L_2), L_{\top}; 1 - \alpha U(L_1) - (1 - \alpha)U(L_2), L_{\perp}]$$

- Applying definition of utility, we get

$$U([\alpha, L_1; (1 - \alpha), L_2]) = \alpha U(L_1) + (1 - \alpha)U(L_2).$$





Decision Analysis

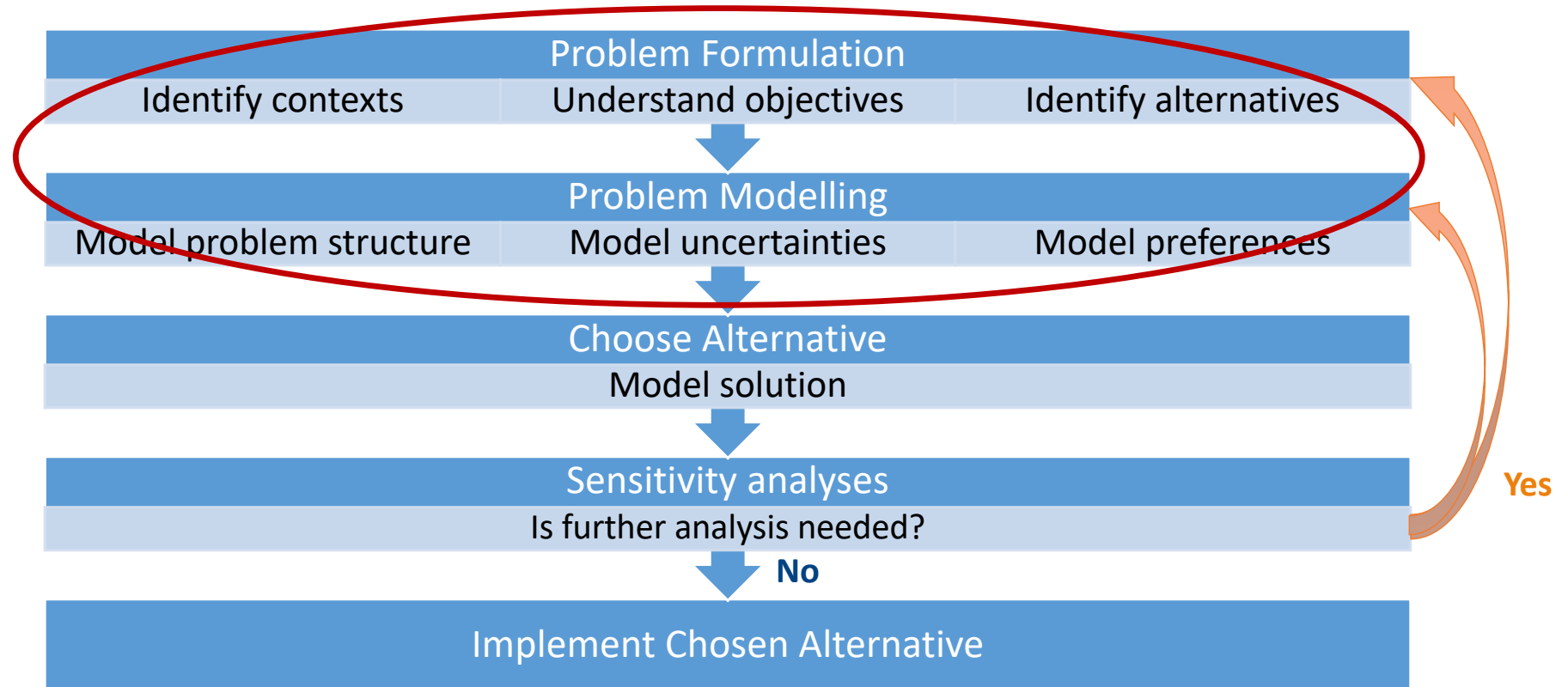
A prescriptive framework for decision making



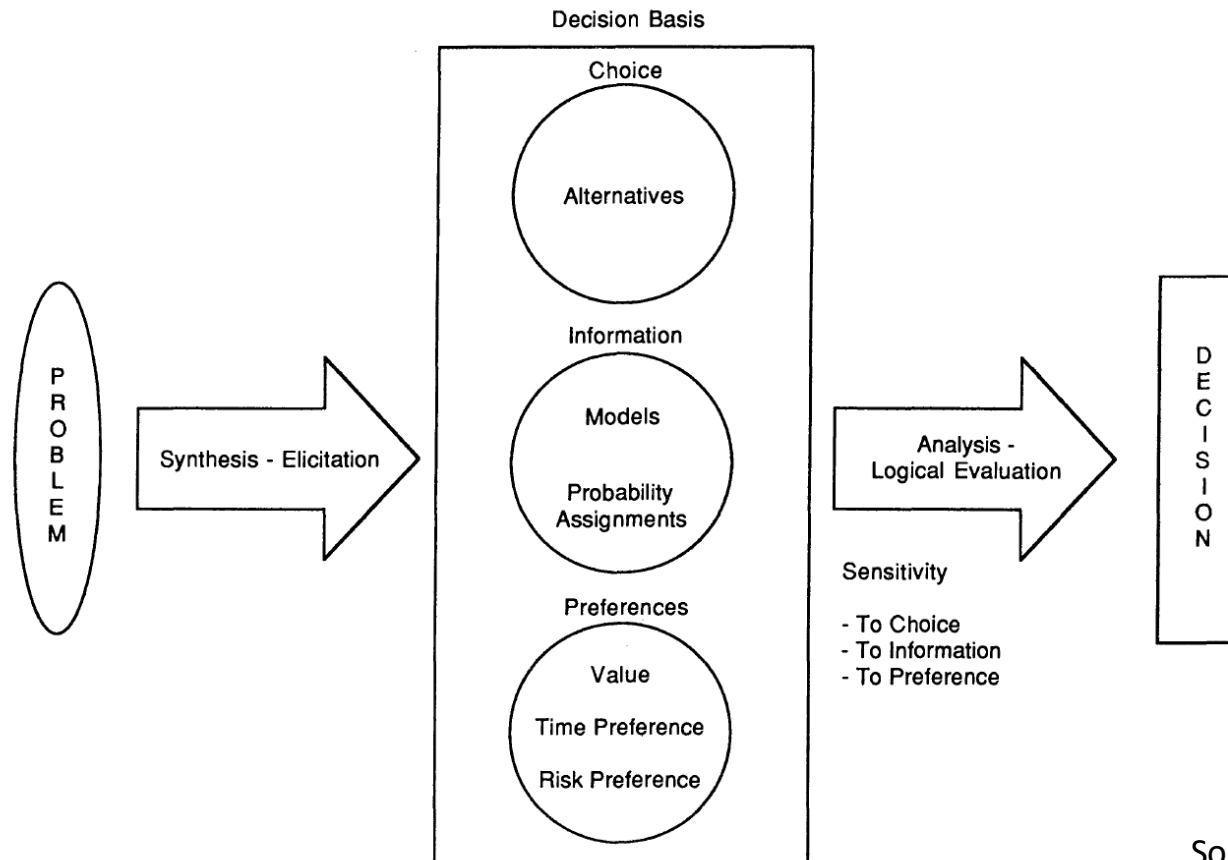
Example: Martian Adventures

- Scenario:
 - Mars Base is considering deploying a new tool for Persy the Mars Rover to collect and analyze Martian rock samples:
 - Two different power sources: Solar Power and Battery Power.
 - Both power designs have different energy consumption rates and development time.
 - Success rate for either design working efficiently in the harsh Martian environment is uncertain.
 - Only one design can be deployed at a time due to payload constraints on the resupply ships.
 - If either design doesn't work optimally once deployed, a new, modified version may be sent in the next resupply mission, but this means waiting longer and it still doesn't guarantee success.
- Two stage decision problem:
 - Decide on which power design to deploy: Solar, Battery, or neither (if risk is too high).
 - If the tool fails, decide whether to send a modified version or to abort the mission objective.

The Decision Analysis Process



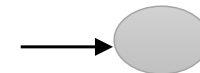
Decision Modeling: Decision Basis Formulation



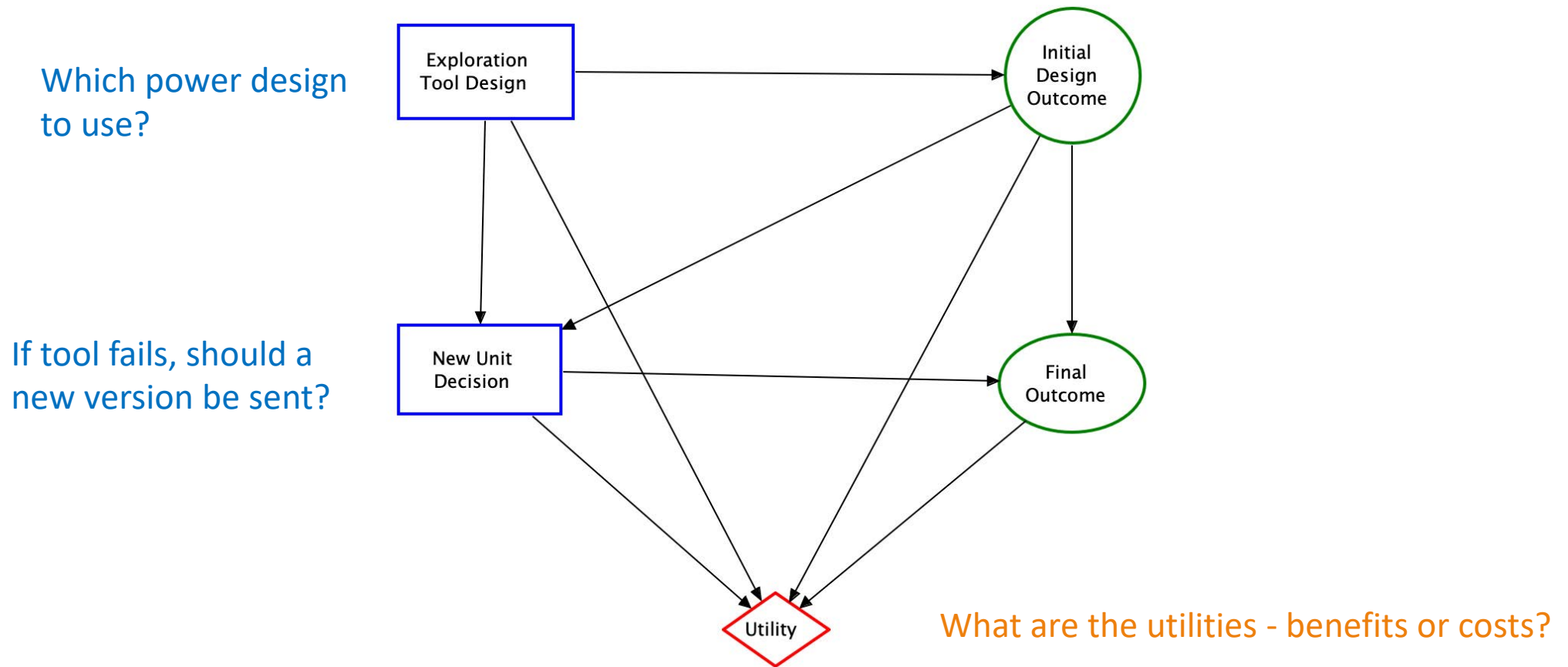
Source: Howard, R. A. 1988

Basic Elements of a Decision Model

- **Decision nodes**
 - Decision points with several choices or alternatives
- **Chance nodes**
 - Uncertain or chance events with several outcomes
- **Value or utility nodes**
 - Deterministic monetary value or utility functions
 - Measure desirability of final objectives
- **Probabilistic dependencies**
 - Represent conditional dependence of chance events
- **Informational dependencies**
 - Represent information available at decision points

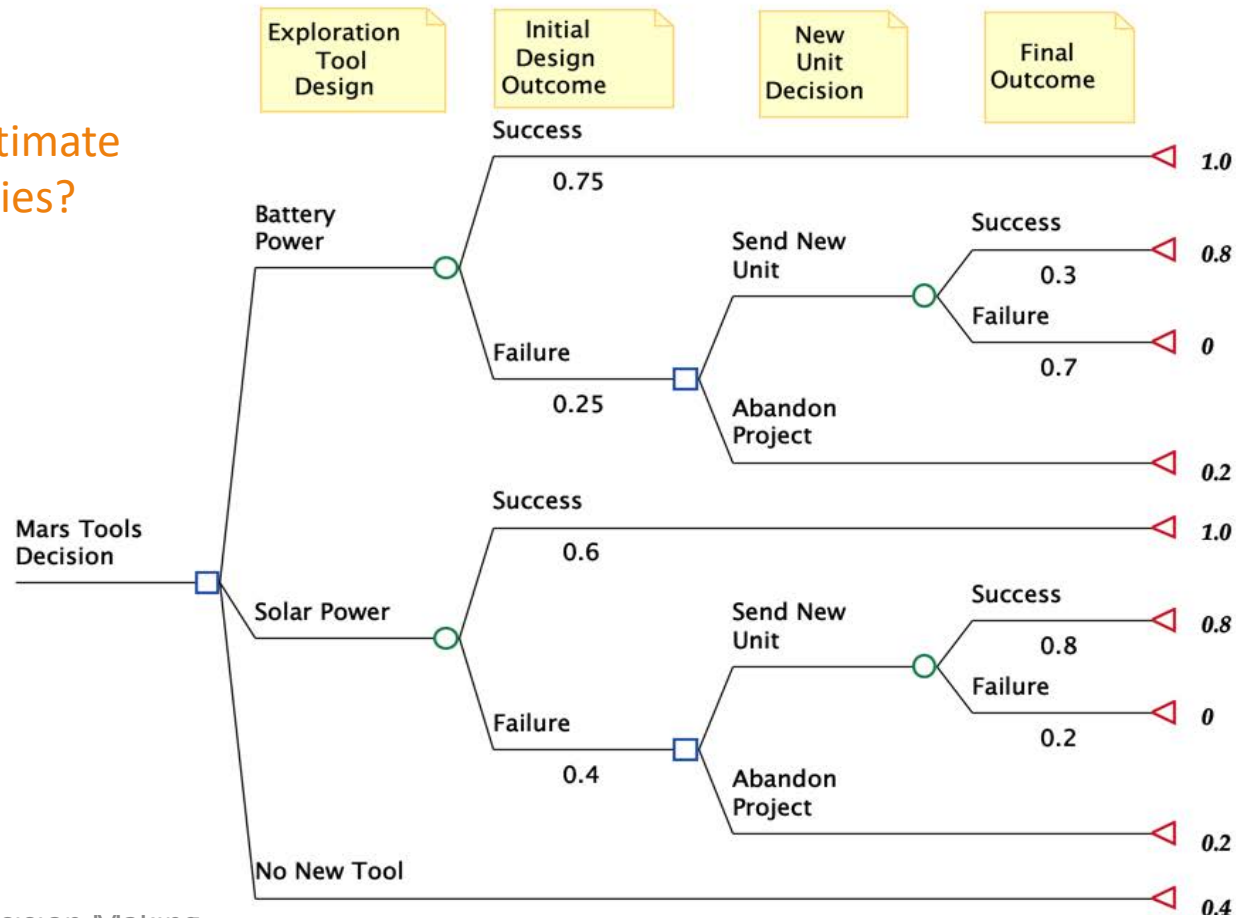


Example: A Decision Model (1)



Example: A Decision Model (2)

How to estimate uncertainties?



How to estimate utilities?



Computing Tools

- Commercial Products – mostly derived from academic research work
 - DPL – Decision Programming Language – Influence diagrams, Decision trees
 - Netica – Bayes nets, Influence diagrams, API engines
 - Treeage Pro – Decision trees, Influence diagrams (limited)
 - Palisade PrecisionTree – Decision trees, Influence diagrams
 - BayesiaLab – Bayes nets, causal analysis
 - Hugin Expert – Bayes nets, Influence diagrams
 - ...
- Commercial Products With FREE academic version
 - [Bayesfusion](#): GeNIe and SMILE - Bayes nets, Influence diagrams, API engines
 - ...
- Also check out survey at:
 - INFORMS Decision Analysis Software Survey
<https://www.informs.org/ORMS-Today/OR-MS-Today-Software-Surveys/Decision-Analysis-Software-Survey>